

DUAL-CORE OPTICAL MOTION TESTBED

MECHANICAL PLATFORM & INTERFACE PCB

Rotary/translation hardware, differential encoder interface, STEP/DIR control, and physical verification

Prepared by	Herder Elektronische Systemen
Report scope	Mechanical architecture, custom interface PCB, fabrication evidence, physical assembly, and verification
Evidence basis	Source code, engineering plots, project photographs, CAD/PCB artifacts, and recorded demonstrations
Revision	4.0 - 5 August 2026

DOCUMENT PURPOSE

This report documents the physical testbed and its custom electrical interface. Firmware, GUI, and measured control results are intentionally contained in the companion firmware report so each document has a distinct customer-facing purpose.

1. Platform Overview

The platform combines a rigid aluminum frame, rotary stage, translation mechanism, optical-table-style upper surface, geared stepper drive, and differential encoder feedback. The custom interface PCB conditions encoder signals and organizes STEP/DIR, control, and power connections between the motion hardware and the STM32H747 target.

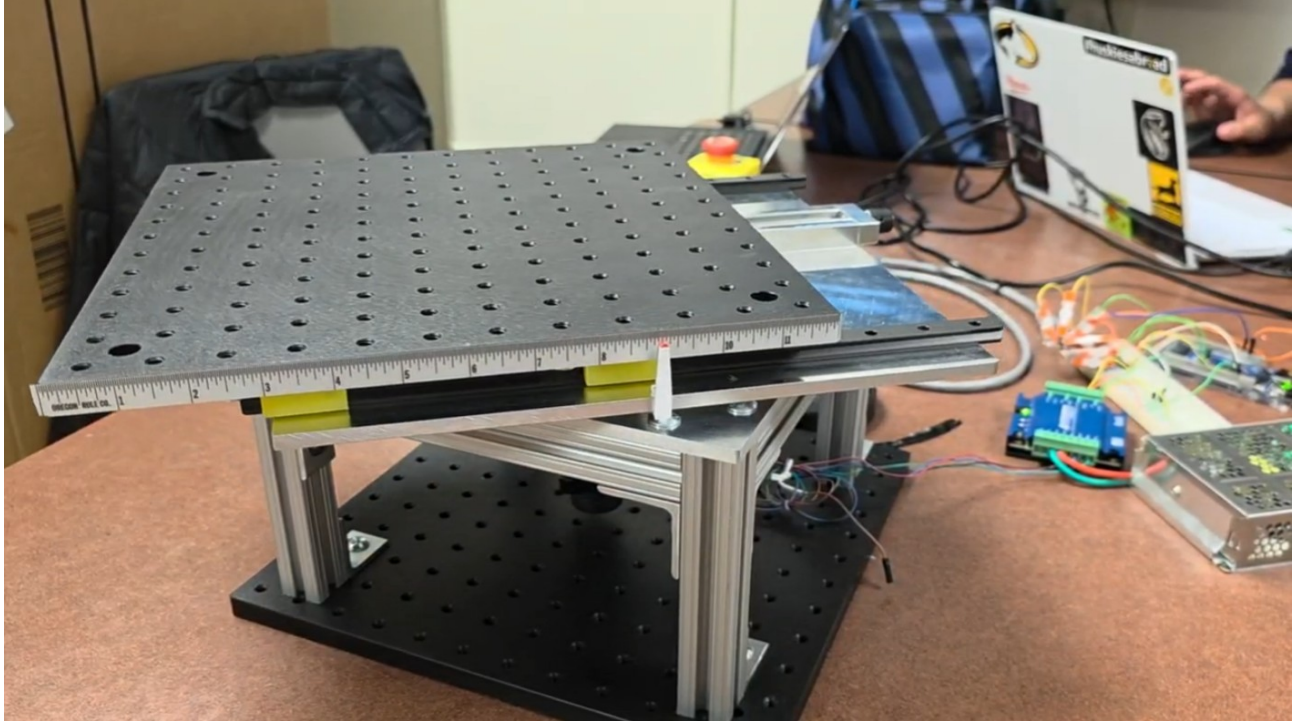


Figure 1. Completed physical testbed with rotary and translation hardware installed.

2. Mechanical Architecture

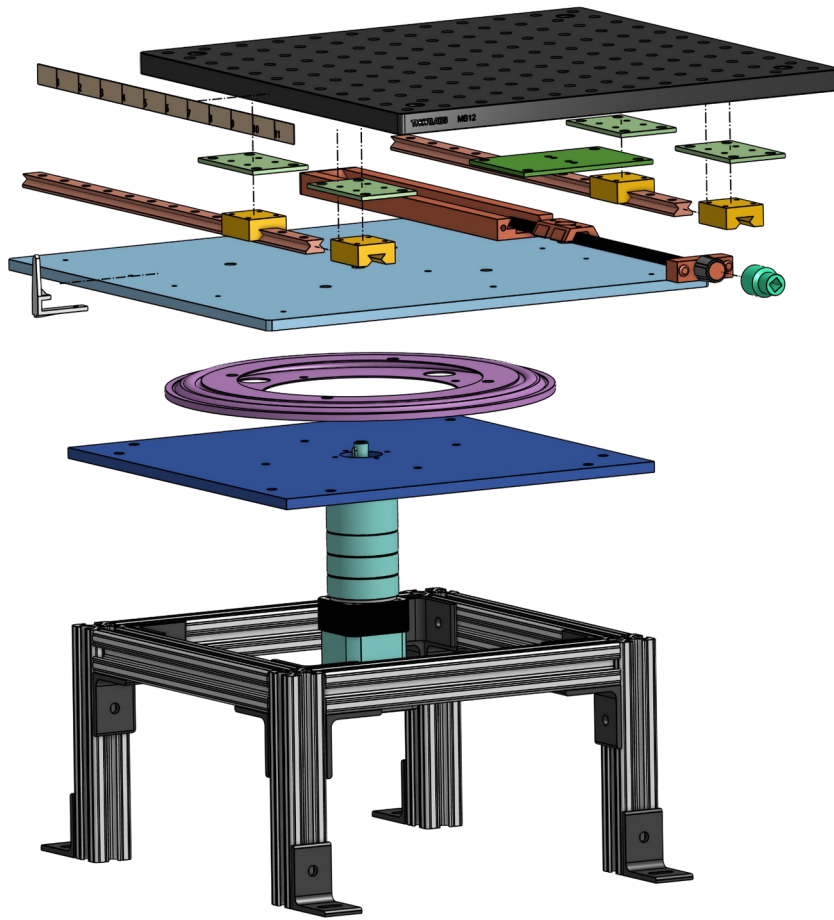


Figure 2. Exploded CAD view showing frame, rotary stage, translation rails, upper plate, and drive components.

- Rotary and translational degrees of freedom are integrated into one bench-scale assembly.
- The frame uses modular aluminum extrusion and a lower base plate for stiffness and serviceability.
- The top plate supports optical fixtures and controlled offset loading.
- The translation mechanism provides more than ± 3 inches of travel in the demonstrated configuration.

3. Fabrication and Assembly Evidence



Figure 3. In-progress assembly showing the frame and mechanical integration.

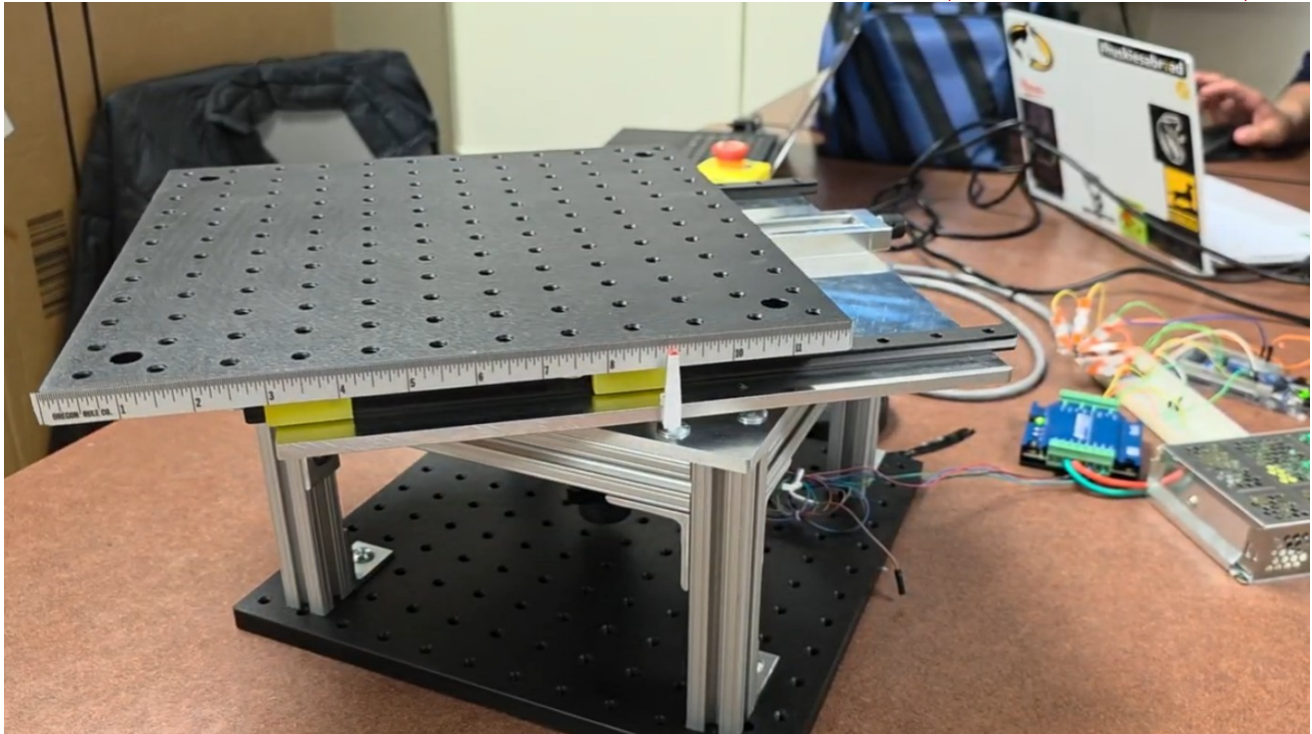


Figure 4. Final assembled testbed configuration.

4. Translation and Alignment Verification

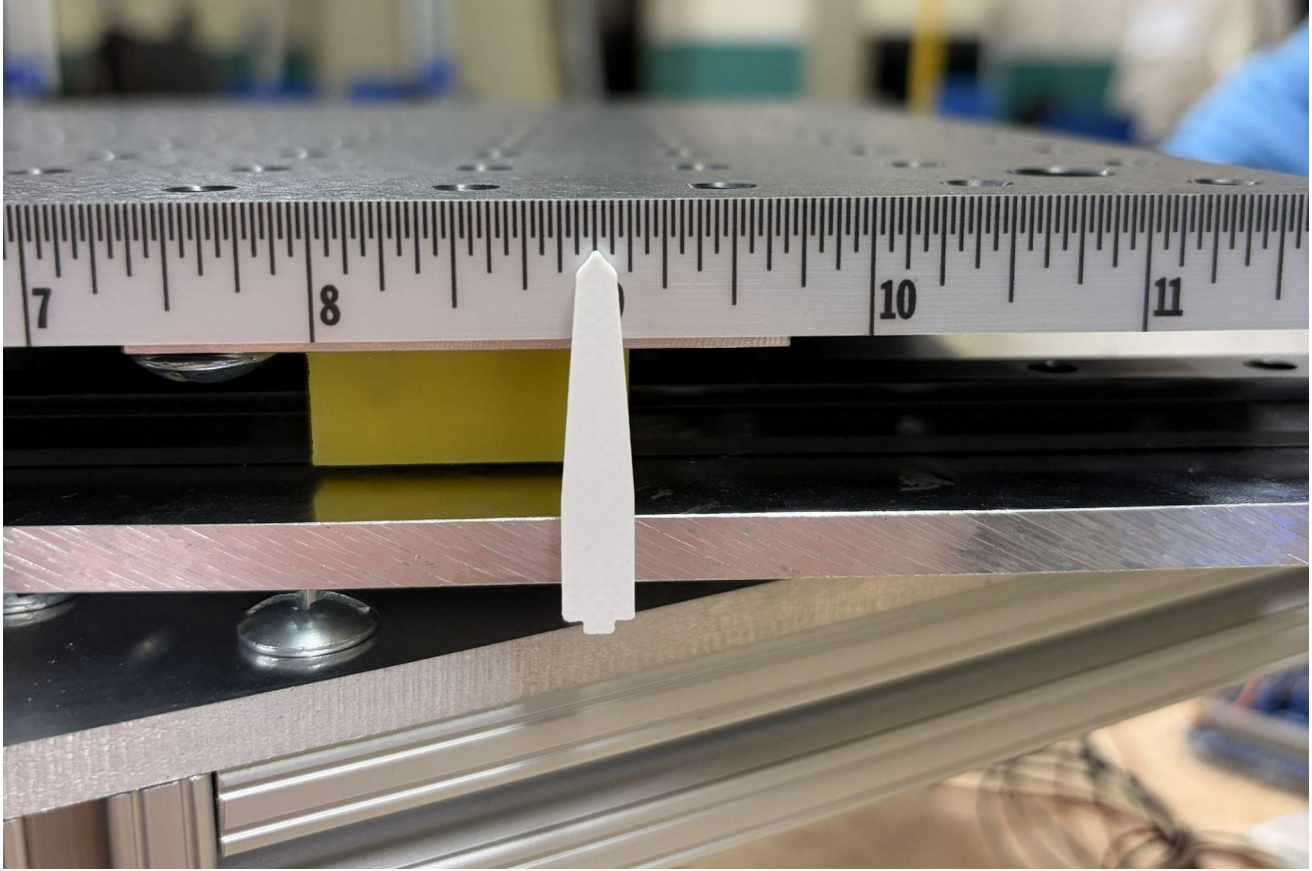


Figure 5. Needle-based translation reference used to observe displacement and alignment.

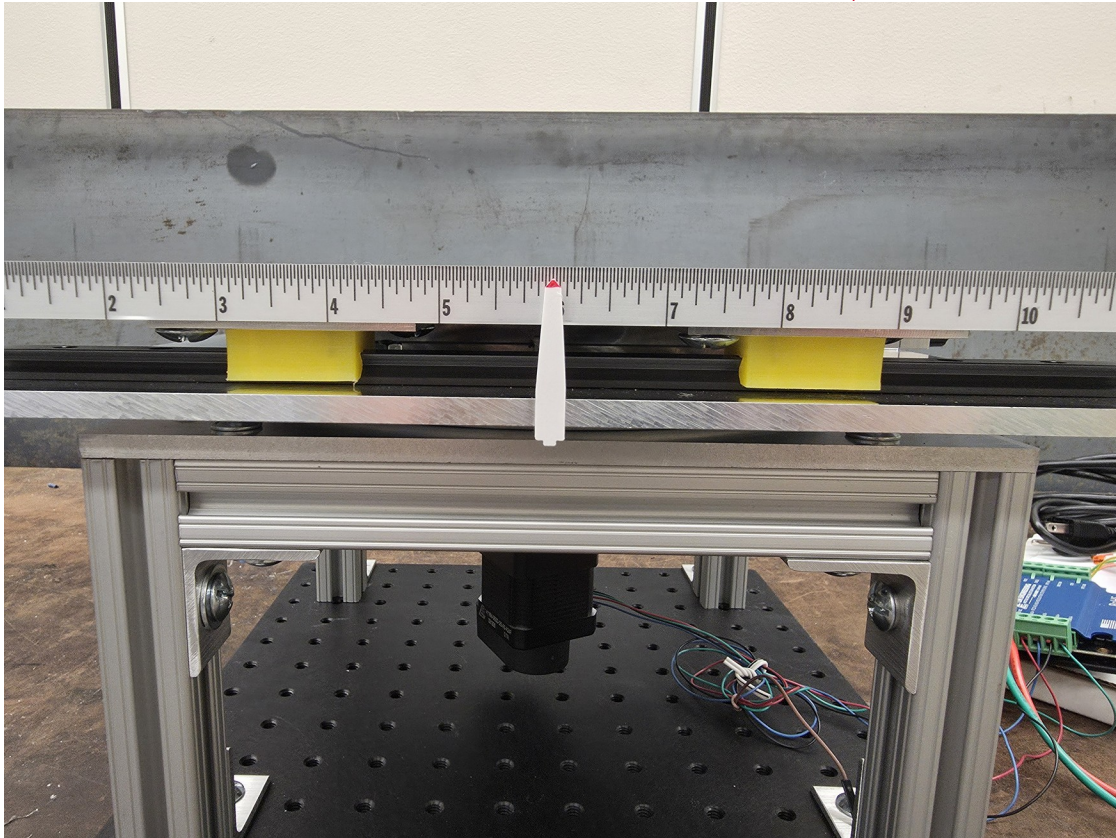


Figure 6. Centered translational precision needle during physical verification.

5. Static Load Demonstration



Figure 7. Forty-pound static offset-load demonstration on the completed platform.

Condition	Observed evidence	Scope
Centered load	Platform supports the applied static mass	Demonstration, not certification
Offset load	Motion hardware remains assembled and functional	Representative bench test
Translation reference	Needle and ruler provide visible displacement reference	Qualitative/bench verification

6. Custom Encoder / STEP-DIR Interface PCB

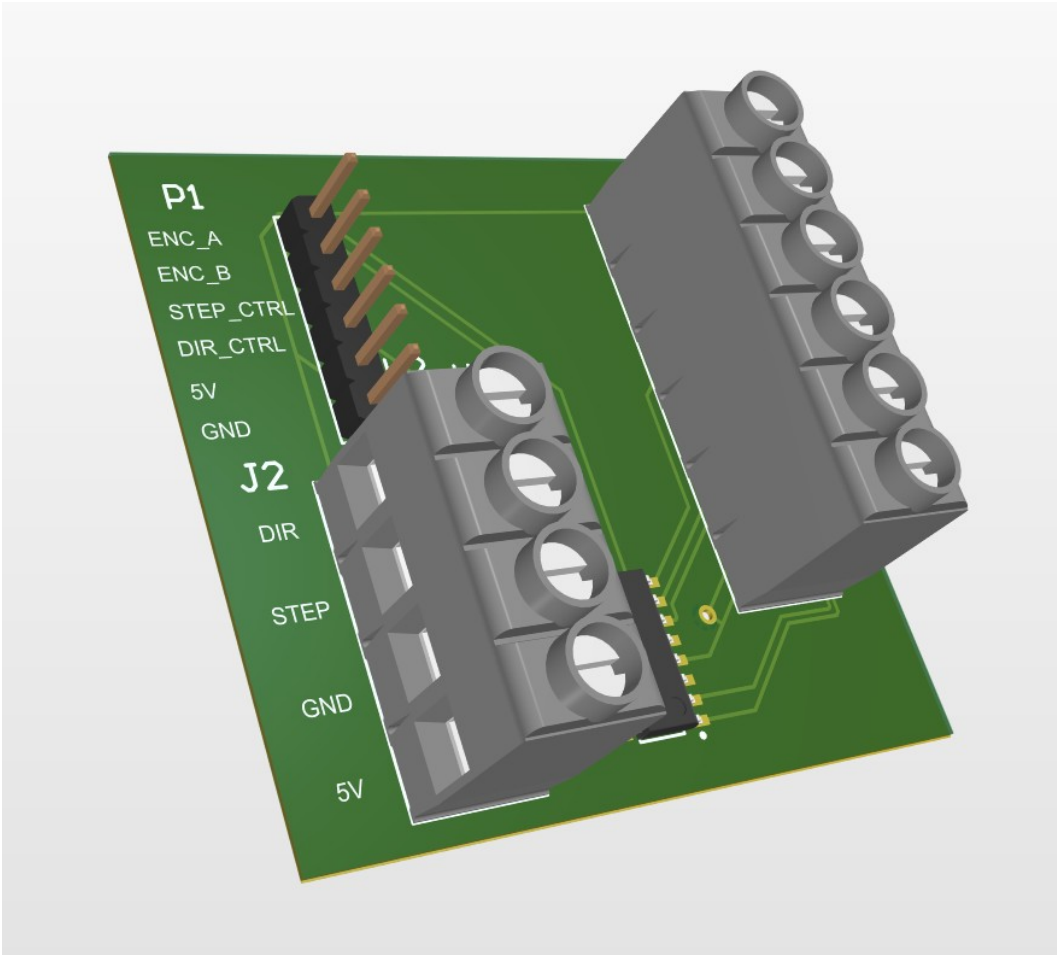


Figure 8. Populated 3D rendering of the custom interface PCB.

Circuit element	Function
AM26LS32 differential receiver	Converts differential encoder channels to logic-level signals
Encoder connector	Routes A/B and related encoder signals
STEP/DIR connector	Routes motion commands and control references
Controller header	Interfaces the board with the STM32H747 target
Terminal blocks	Provide field-wiring access for encoder, control, power, and ground

7. Schematic and Signal Routing

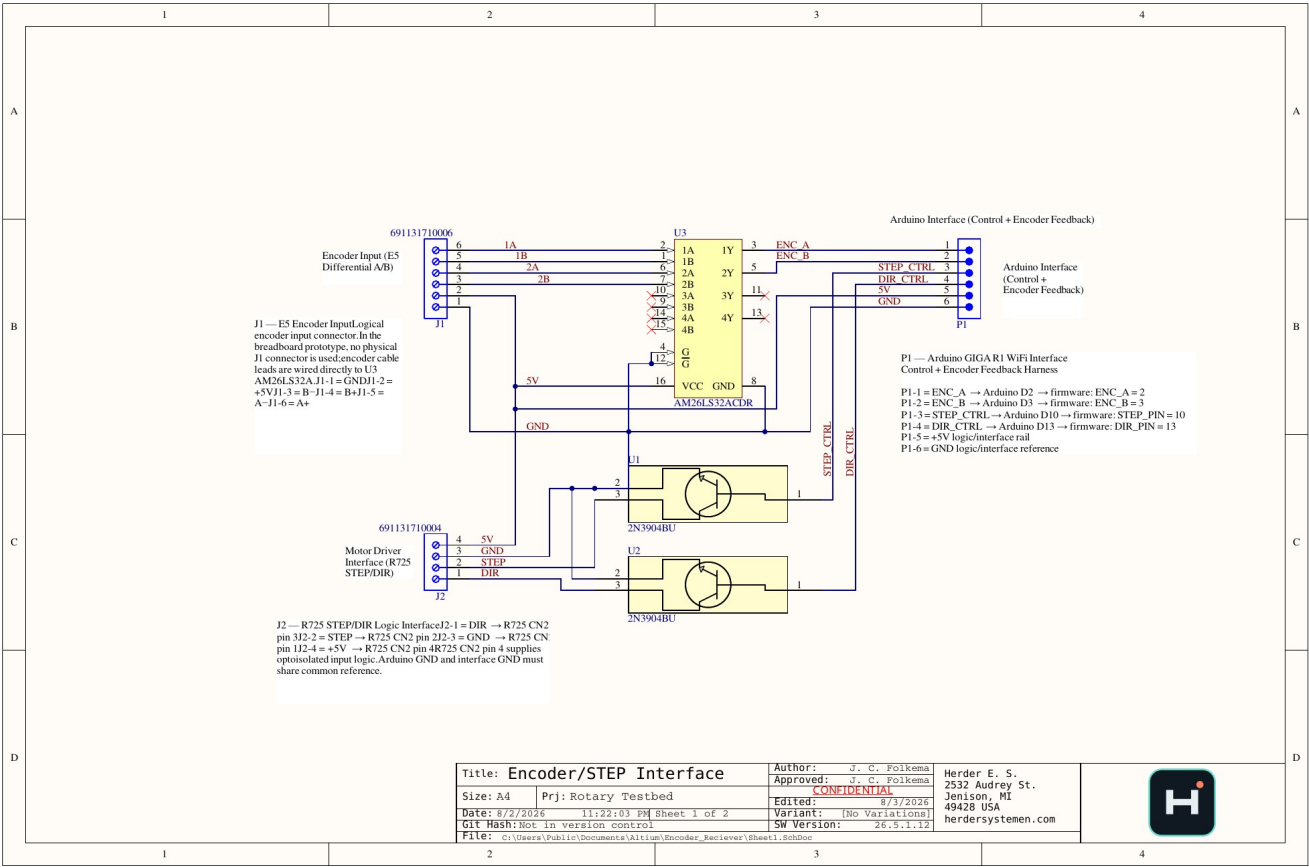


Figure 9. Encoder and STEP/DIR interface schematic.

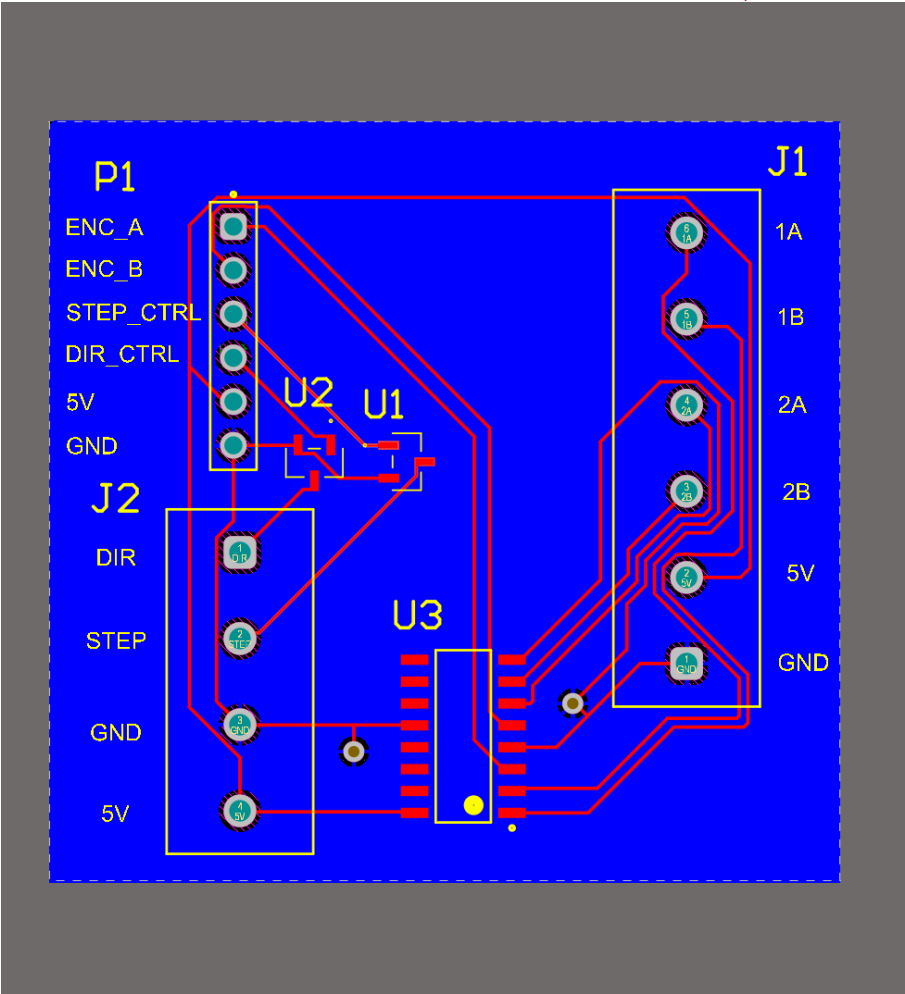


Figure 10. Routed signal layer with component placement and connector organization.

8. Dedicated Ground Plane

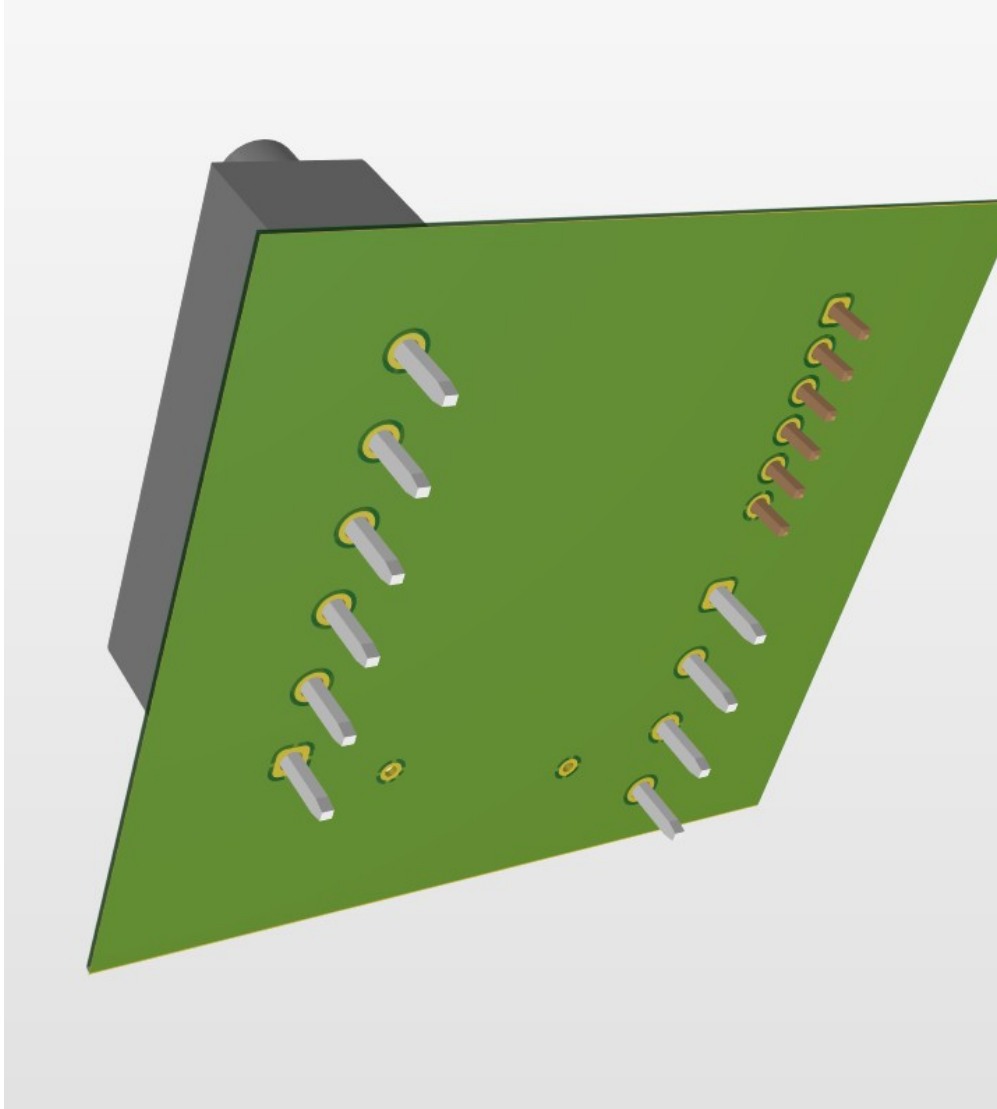


Figure 11. Dedicated solid-copper ground plane on the opposite PCB layer.

The dedicated ground plane provides a continuous return reference beneath the routed signal layer. This is particularly important for differential encoder reception, controller reference stability, and minimizing fragmented return paths across the compact interface board.

9. PCB Assembly Views

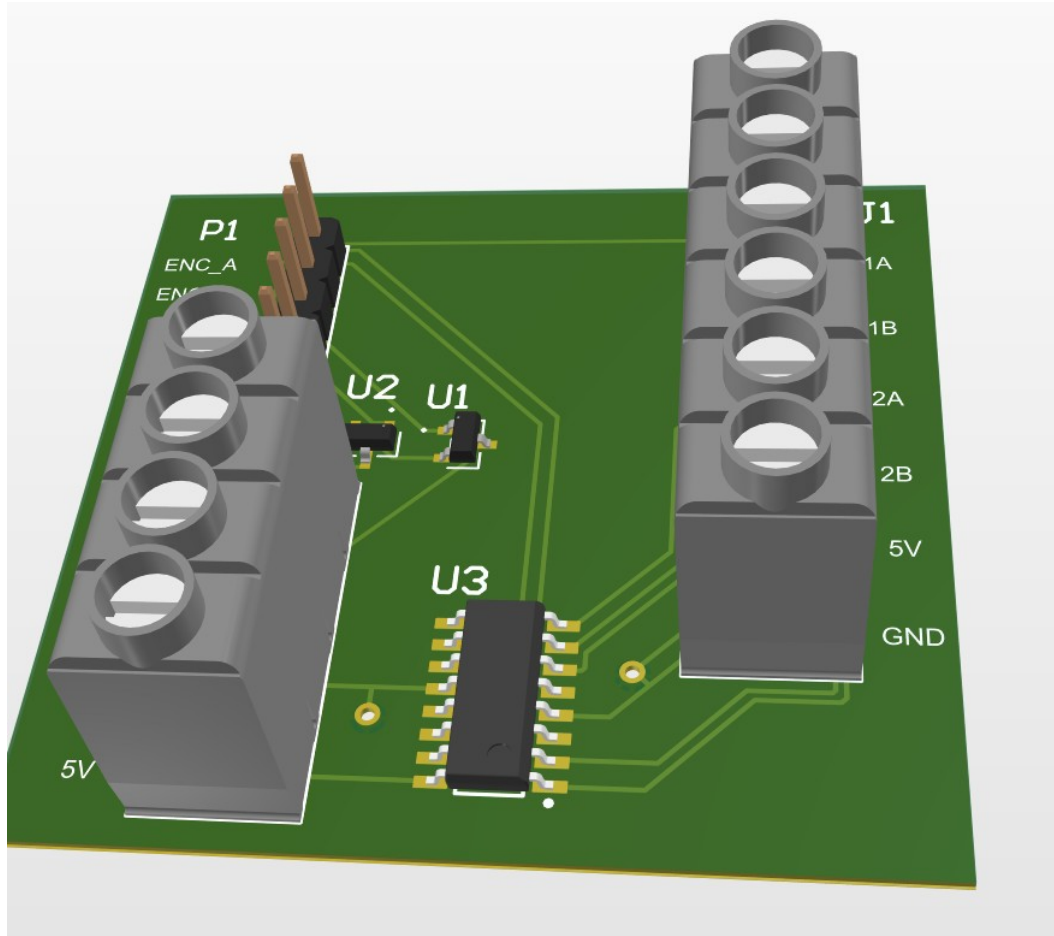


Figure 12. Angled PCB assembly rendering showing terminal blocks, receiver IC, and headers.

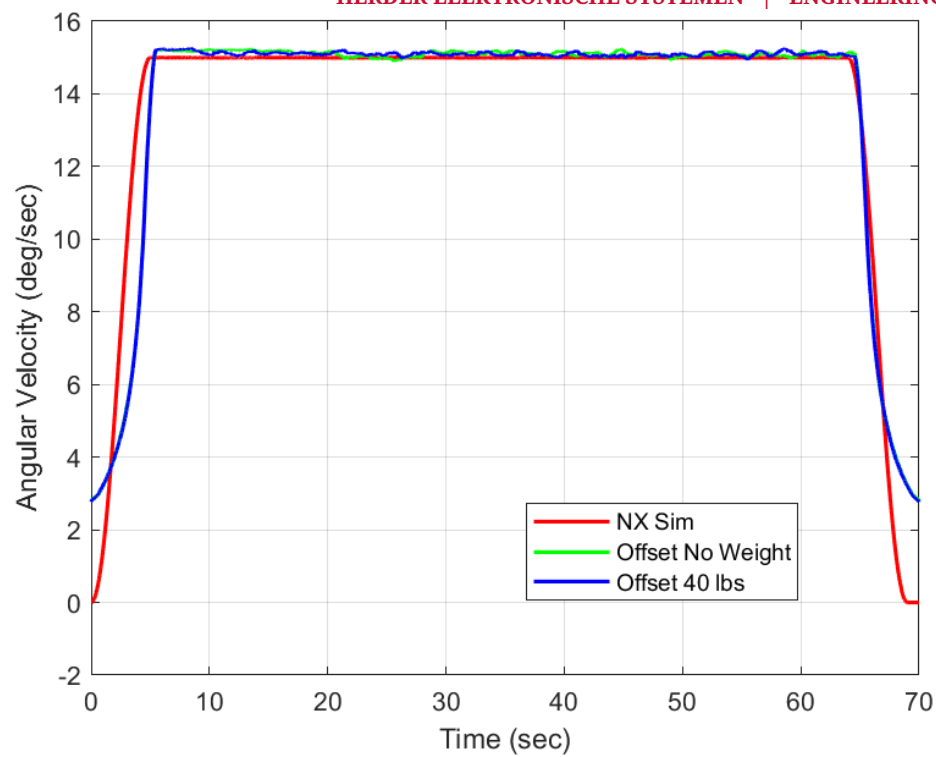


Figure 13. Alternate PCB assembly view.

10. Dynamic Verification Evidence

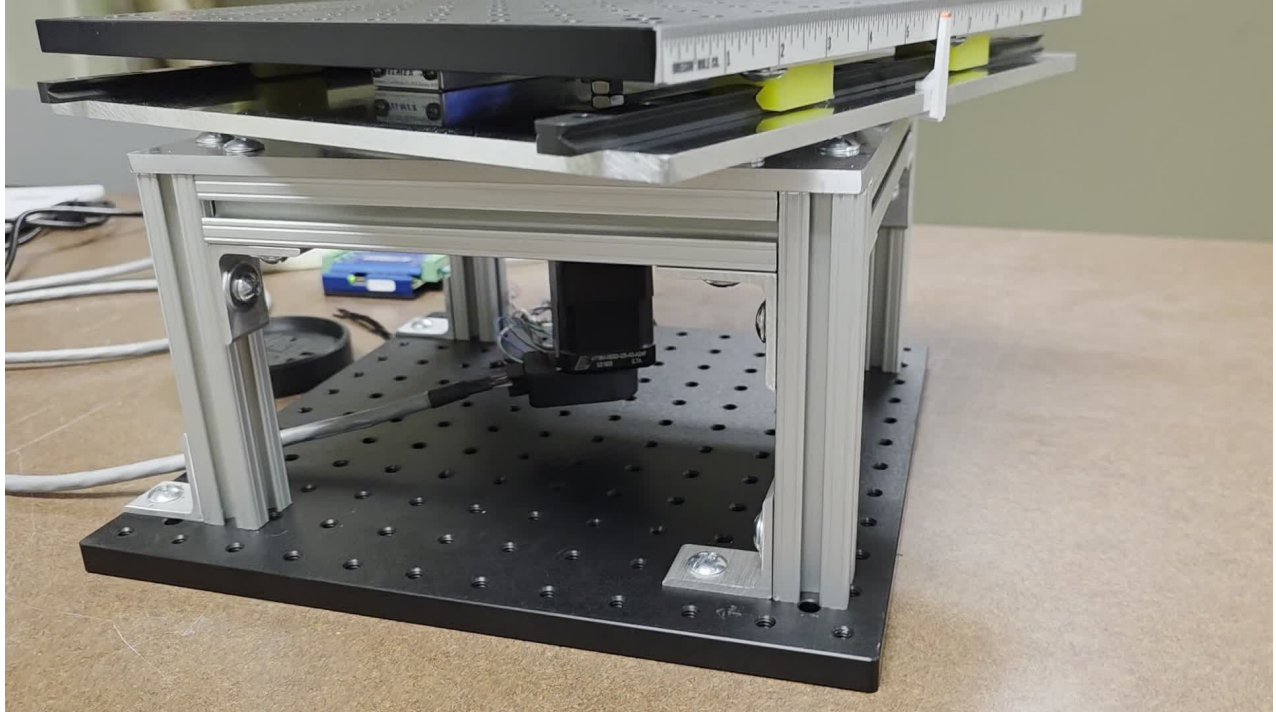


Figure 14. Frame from the integrated project-solution demonstration video.

Solution1	
Time	5.880000
Step	588

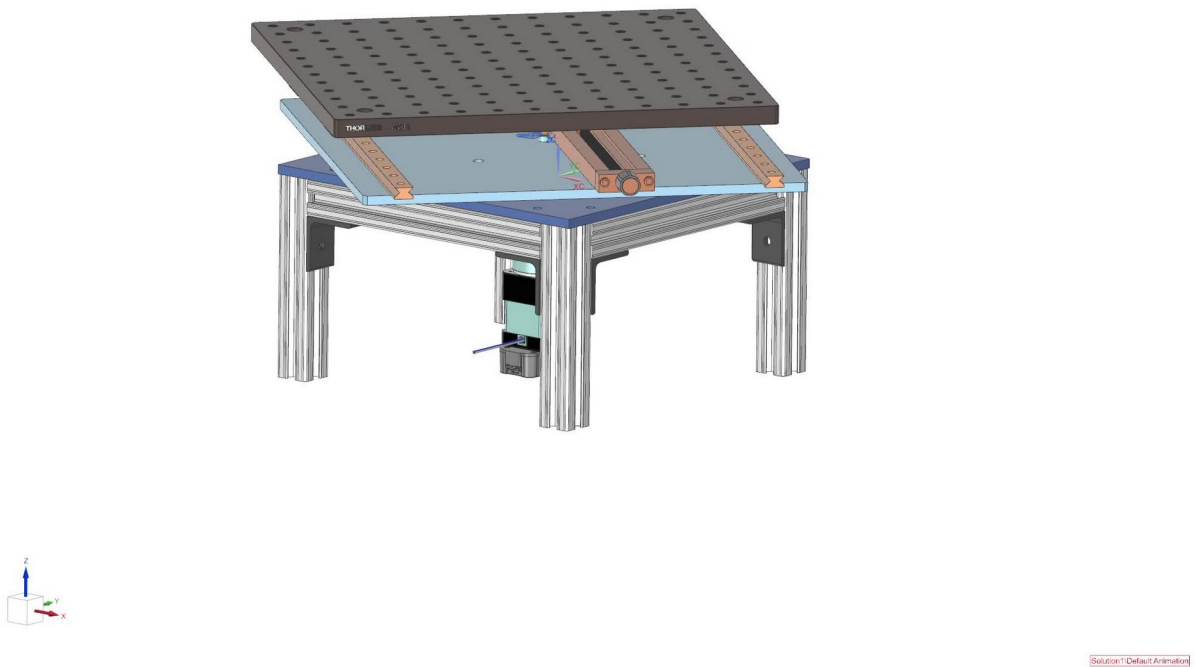


Figure 15. Frame from the NX model animation used for mechanism review.

Recorded demonstrations include the integrated project solution video, an NX model animation, centered and offset load videos, and motion demonstrations. The companion firmware report contains the quantitative control and telemetry results; this report limits its conclusions to physical architecture and observed bench verification.

11. Limitations and Document Control

Item	Controlled statement
Prototype manufacture	No formal shock, vibration, thermal, EMC, or long-duration certification is claimed.
Electrical power	Idle, steady-state, and peak system power remain to be measured.
Fabrication release	Editable board files and the final manufacturing package remain controlled.
Metrology	Motion-platform characterization does not certify downstream optical performance.
Load evidence	The 40 lb result is a static demonstration, not a rated payload certification.

Attribution

Herder Elektronische Systemen developed the public-facing engineering package and documents the platform architecture, custom encoder/STEP-DIR PCB, physical verification evidence, and release controls represented here. No third-party affiliation or endorsement is claimed.

Revision record

Revision	Date	Description
4.0	2026-08-05	Rebuilt from scratch with restored mechanical, schematic, routed-copper, ground-plane, PCB-render, and physical-verification imagery.